

Java

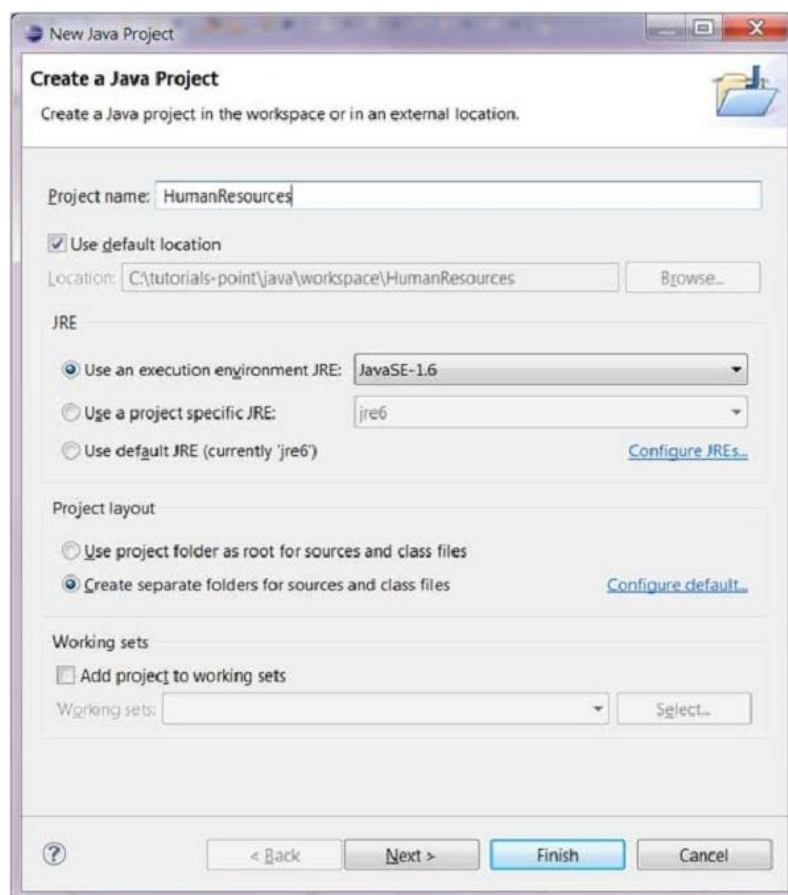
Create Java Project :-

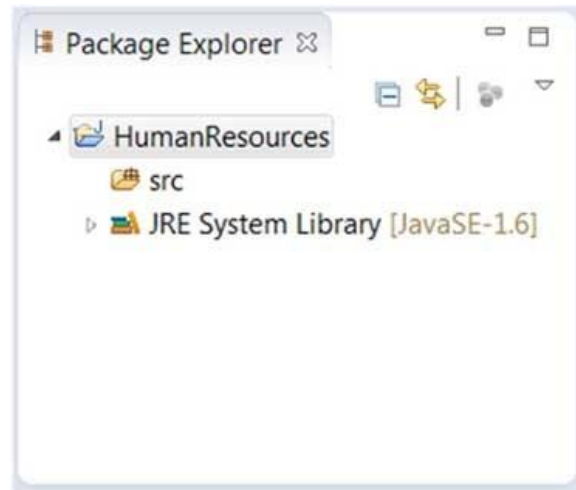
Steps to create a project in Eclipse :-

1. Open Eclipse → Select workspace.
2. File → New → Java Project → Name project → Finish.
3. Right-click src → New → Package → Name it.
4. Right-click package → New → Class → Name → check main method → Finish.
5. Write code:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, Eclipse!");  
    }  
}
```

6. Run As → Java Application → Output in console.





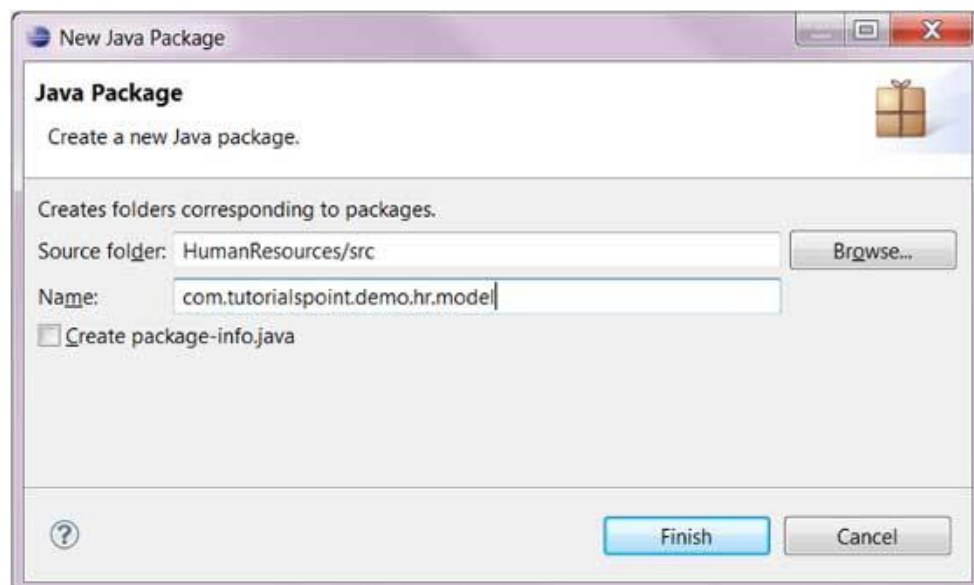
Create Java Package :-

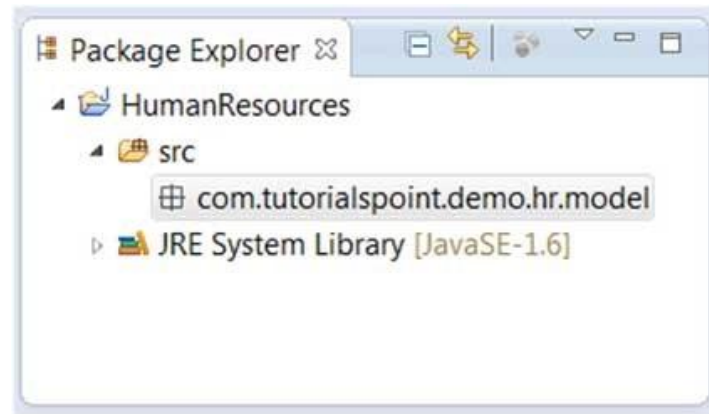
What is Java Package ?

- It's like a folder in your computer.
- Helps in code organization, reusability, and avoiding name conflicts.
- Two types:
 1. Built-in packages - java.util, java.io, etc.
 2. User-defined packages - Created by programmers.

Steps to create a package in Eclipse :-

1. Open your Java Project in Eclipse.
2. Right-click on the src folder → New → Package.
3. Enter a package name (e.g., com.example).
4. Click Finish.





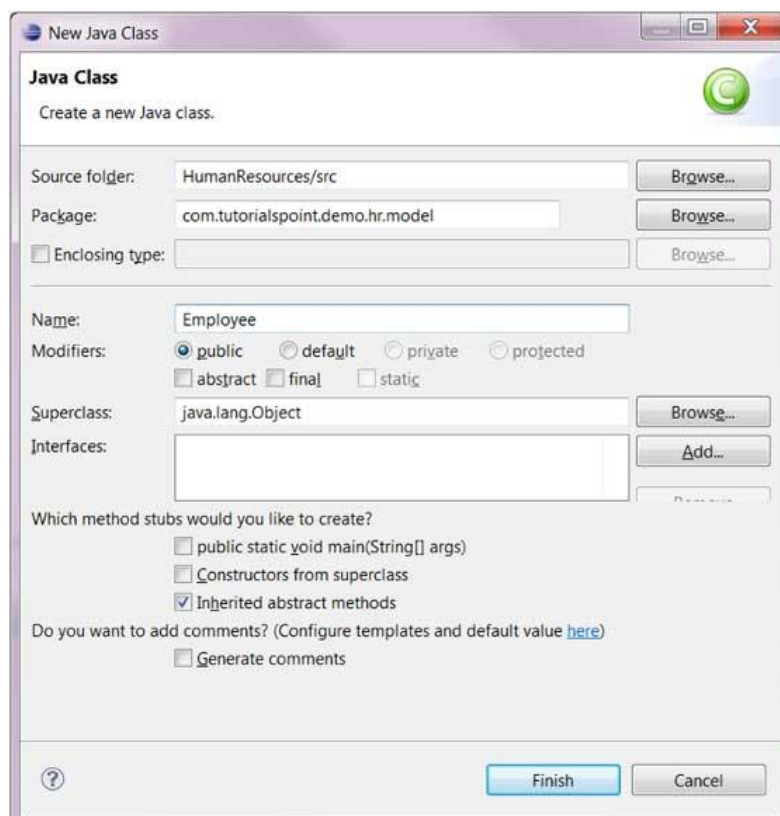
Create Java Class :-

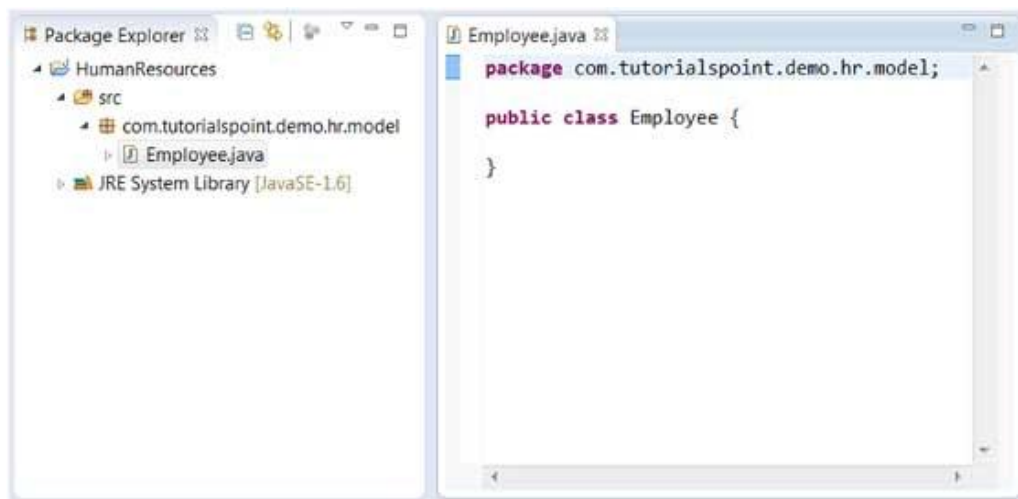
What is Java Class ?

- A class is a user-defined data type.
- It groups variables and methods into a single unit.
- Objects are created from classes.

Steps to create a class in Eclipse :-

1. In your project, right-click on the package (or src if no package).
2. Select New → Class.
3. Enter a Class Name (e.g., HelloWorld).
4. (Optional) Tick public static void main(String[] args) to auto-add main method.
5. Click Finish.





Arithmetic Operation :-

- Arithmetic operators are used for basic math calculations.
- Operators:
 - + → Addition
 - - → Subtraction
 - → Multiplication
 - / → Division (quotient)
 - % → Modulus (remainder)
- Work on numeric data types (int, float, double, etc.).
- Precedence: *, /, % > +, -.
- Integer division gives whole number, floating-point gives decimal.

Operators	Result
+	Addition of two numbers
-	Subtraction of two numbers
*	Multiplication of two numbers
/	Division of two numbers
%	(Modulus Operator)Divides two numbers and returns the remainder

Example –

```
package Basic_Concept;
```

```
public class arithmetic_operation {  
    public static void main(String [] args)  
    {  
        System.out.println("5+3");  
        System.out.println("Addition is:"+(5+3));  
        System.out.println("Substraction is:"+(5-3));  
        System.out.println("Multiplication is:"+(5*3));  
        System.out.println("Division is:"+(6/2));  
    }  
}
```

Output –

5+3

Addition is:8

Substraction is:2

Multiplication is:15

Division is:3